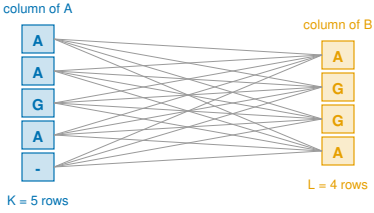
MULTIZ 11.4 scores each cell from column counts: K × L becomes K + L

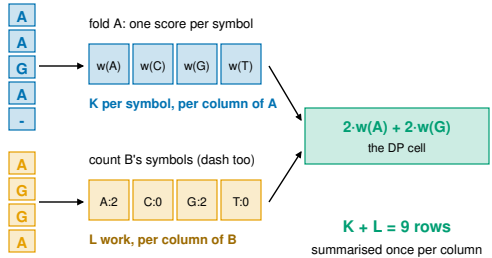
A 11.3: every row pair, in every cell



K × L = 20 pair scores

for every DP cell, four times over
(the substitution sum and three gap sums)

B 11.4: summarise each column once, then combine

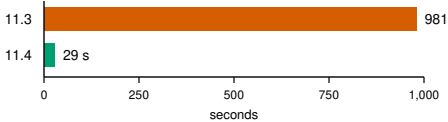


K + L = 9 rows

summarised once per column
then a short dot product per cell

Gap sums are counted the same way. A row's gap cost depends only on whether it holds a dash in this column and in the previous one, so each row falls into one of 4 dash classes. All terms are integers: the totals equal the pair sums exactly.

C One deep merge, as recorded in multiz#28



Blocks of about K = 35 rows against L = 43:

K × L = 1,505 row pairs per cell

K + L = 78 rows summarised per column

33.8× faster on this merge,

identical output (single upstream runs)